

Bangladesh University of Engineering and Technology (BUET)
Department of Industrial and Production Engineering (IPE)



Course No: IPE 6112
Course Title: Probabilistic Methods in Engineering Design

**Preference-Driven MCDA Ranking with Generalized-Gaussian Desirability
and CWA Aggregation**

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Abstract

An end-to-end MCDA-inspired preference-learning ranking pipeline was implemented to examine whether non-monotonic desirability shaping can provide an interpretable alternative to monotonic utility models in settings where simple “more is better” assumptions are not appropriate. Criterion values were standardized and mapped to non-monotonic generalized-Gaussian desirability functions, where deviations from an ideal region were penalized and tolerance was controlled by learned sharpening parameters. Desirability-weighted information values were aggregated into global utilities using a compensative weighted-averaging operator, and the aggregation attitude was governed by a learned parameter that allowed AND-like to OR-like behavior to be expressed. Pairwise preferences were generated from ordinal ground-truth ranks, and model parameters were estimated by maximum-likelihood learning under a Bradley–Terry formulation. Evaluation was conducted on three UCI choice sets (Auto MPG, Car Evaluation, and Computer Hardware) using 100 randomized train–test trials, with pairwise ranking accuracy used as the primary metric. Mean accuracies of 0.8190 ± 0.0141 , 0.8958 ± 0.0097 , and 0.8508 ± 0.0292 were obtained under the full model, indicating stable generalization under resampling. Ablation studies showed that performance was degraded when desirability shaping or learned weights were removed, while a monotonic MAUT baseline remained competitive when ordinal labels were induced from a single monotonic target.

Keywords: Multi-Criteria Decision Analysis (MCDA), Preference Learning, Generalized-Gaussian Desirability, Compensative Weighted Averaging (CWA), Bradley–Terry Ranking Model.

1. Introduction

In practical choice problems, many alternatives are compared across multiple—often interacting—criteria, and large volumes of information are processed by a decision-maker (DM). MCDA methods are used to structure this evaluation; however, many commonly used models are built on monotonic preference assumptions (e.g., “more is always better” or “less is always better”). In real settings, these assumptions are often violated because criteria are conflicting and closely interrelated. As a result, extreme performance on a single criterion is frequently not preferred when it is achieved at the expense of other criteria (e.g., the least-cost

option is associated with minimal features). Balanced criterion profiles are therefore often favored over extremes.

In this study, monotonic utility is replaced by a parametric bell-shaped desirability function in which desirability is maximized at an optimal criterion value. Values near the optimum are assigned higher desirability, while deviations on either side are penalized. The location of the optimum is defined in a choice-set-sensitive manner, since it is derived from the criterion’s distribution across the available alternatives. Consequently, the method is adapted to the specific context of the current choice set rather than being fixed a priori. In addition, the desirability curve is made individualizable through a DM-specific attitudinal parameter that controls the rate at which desirability declines away from the optimum. With greater rigidity, a sharper peak is produced, and a steeper penalty is imposed for non-optimal values.

Two main contributions are made in this project. First, an MCDA-inspired ranking procedure is presented in which context-dependent, non-monotonic desirability is operationalized through a symmetric and sharpenable desirability curve, so that balanced alternatives can be preferred over lopsided extremes. Second, the resulting ranking model is evaluated within a machine learning (ML) framework, and empirical performance is demonstrated on multiple datasets with differing target structures.

2. Literature Review

Multi-criteria decision aiding (MCDA) methods are commonly distinguished by how criterion-level preferences are represented and how those preferences are aggregated into an overall evaluation. Classical multi-attribute utility approaches provide an intuitive criterion-by-criterion utility interpretation, but they typically rely on monotonic preference assumptions and fixed compensation behavior, which may be restrictive when desirable performance is concentrated around a moderate or context-dependent range rather than at an extreme value (Ittersum & Pennings, 2012; Pham et al., 2015; Wallenius et al., 2008; Valkenhoef & Tervonen, 2016). In parallel, outranking and interactive MCDA frameworks offer richer preference elicitation mechanisms, but they are less directly aligned with the present objective of learning a compact, criterion-wise desirability-and-aggregation model from ordinal comparisons (Brans, 1982; Xanthopoulos et al., 2000).

Table 1: MCDA Literature Landscape: Utility/Desirability Elicitation and Aggregation Operators

Theme	Representative works cited in the paper	What the line of work does	Typical strength	Key limitation (why this paper says it's not enough)
MCDA family: MAUT	Ittersum & Pennings (2012); Pham et al. (2015); Wallenius et al. (2008)	Assigns utility/desirability per criterion and aggregates into overall utility	Intuitive, widely used; clear “utility per attribute” story	Needs <i>DM-specific</i> utility shaping + DM-specific aggregation behavior (compensation)
MCDA family: Outranking	Brans (1982)	Compares alternatives via outranking relations (prefer/incomparable/indifferent)	Handles incomparability explicitly	Not the paper’s focus; doesn’t directly model “utility shaping + compensation” in the MAUT sense
MCDA family: Interactive	Xanthopoulos et al. (2000)	Uses analyst–DM interaction to infer preferences	Can adapt to DM	Interaction burden; still needs a compact preference model
Utility elicitation via discrete points + interpolation	van der Pol et al. (2014)	Elicit utility at a few points, then interpolate a curve	Simple to implement	Coarse, not robust; limited realism of desirability shaping
Revealed-preference utility elicitation	Abdellaoui et al. (2013); Tsetlin & Winkler (2012)	Elicit utility from DM’s revealed choices/preferences	Behaviorally grounded	Requires extensive questionnaires / heavy elicitation → limited practicality
Optimization-based utility elicitation (as MCDA)	Cui et al. (2024); Lamontagne et al. (2023); Qu et al. (2023)	Learns decision parameters through optimization formulations	Often intuitive and principled	Paper argues limited for <i>criterion-level desirability modeling</i> (the specific need here)
Aggregation: simple additive	Chen et al. (2022); Langhans et al. (2014); Zhou et al. (2024)	Sum/average desirabilities (often implicitly assumes fixed tradeoff)	Easy, scalable	Assumes a fixed compensation behavior; too rigid for diverse DM attitudes

Theme	Representative works cited in the paper	What the line of work does	Typical strength	Key limitation (why this paper says it's not enough)
Aggregation: weighted averaging	Valkenhoef & Tervonen (2016)	Adds explicit criterion weights	Adds criterion importance	Still “representative averaging”; compensation style remains limited
Aggregation: OWA (Ordered Weighted Averaging)	Aggarwal (2015); Reimann et al. (2017)	Aggregates after sorting criterion utilities; OWA weights capture attitude	Can represent AND-like $\leftrightarrow$ OR-like behavior	Still yields a fixed representative value; paper targets more flexible compensation control
Aggregation: Choquet / attitudinal Choquet	Aggarwal (2018); Aggarwal & Fallah Tehrani (2019)	Models interactions among criteria + attitude	Captures interaction effects beyond additive	Paper notes baselines may not model “perceived feature values” governing choice (and/or can be heavy to fit)

Table 1 summarizes the main MCDA streams most relevant to the present study. The literature shows that prior work has addressed either utility elicitation or aggregation flexibility, but rarely both in a form that remains simple, interpretable, and suitable for learning from pairwise ranking data. Point-based interpolation and revealed-preference utility elicitation can provide behaviorally grounded preference representations, yet they often require more direct elicitation effort than is practical in many settings (van der Pol et al., 2014; Abdellaoui et al., 2013; Tsetlin & Winkler, 2012). Likewise, additive, weighted, OWA, and attitudinal Choquet formulations provide different degrees of compensation control, but they do not directly impose a criterion-wise sweet-spot desirability structure in which both under-performance and over-performance can be penalized around a reference region (Chen et al., 2022; Langhans et al., 2014; Zhou et al., 2024; Aggarwal, 2015; Reimann et al., 2017; Aggarwal, 2018; Aggarwal & Fallah Tehrani, 2019). Relative to these lines of work, the present model introduces a bell-shaped desirability layer before aggregation, so that local criterion preference shape and global compensation attitude are learned as distinct components rather than being conflated into a single weighted summary.

Table 2: Benchmark Methods Used in the Paper: Preference Learning and MCDA Baselines

Method	Citation in paper	Core idea	What it captures well	Main drawback vs. the proposed model
RankSVM	Herbrich et al. (2000) Herbrich et al. (2000)	Large-margin learning from pairwise preferences	Strong, standard learning-to-rank baseline	Not an interpretable “utility + compensation” model of the DM; mostly predictive
RBF Kernel model	Han et al. (2012); Claesen et al. (2014)	Nonlinear kernel to fit preference patterns	Nonlinear dependencies	Kernel choice/ σ tuning; less transparent DM choice model
Linear utility (weighted sum)	Stanimirovic et al. (2011)	Weighted sum across criteria	Simple, interpretable	Too restrictive; cannot represent complex desirability shapes/attitudes
Distance from Ideal Solution	Chakraborty et al. (2023)	Learn an “ideal” vector and rank by closeness	Intuitive geometry; can mimic some interaction via distance weighting	Still a proxy; doesn’t explicitly model criterion-level desirability + DM compensation
Tchebycheff utility/distance	Lin et al. (2024)	Generalized norm distance to ideal with tunable (p)	Handles worst-case emphasis as $(p \rightarrow \infty)$	Norm choice is still a limited family of behaviors
TOPSIS	Hwang & Yoon (1981)	Rank by near to PIS and far from NIS	Well-known MCDA ranking workhorse	Reference-point heuristic; not a learned desirability function per criterion
Attitudinal Choquet Integral (ACI)	Aggarwal (2018); Aggarwal & Fallah Tehrani (2019)	Learn fuzzy measure + attitude to model interactions	Strong when criteria interact + attitude matters	Paper argues it can miss “perceived feature value” effects; also heavier structure/constraints
Preference Learning overview	Fürnkranz et al., (2014)	Broad preference learning framing	Positions baselines in PL context	Not a method itself; used to situate PL as a field

From a predictive perspective, the benchmark methods in Table 2 span both standard preference-learning models and widely used MCDA ranking heuristics. RankSVM and RBF-kernel methods are strong ranking baselines because they can fit pairwise preference structure with linear-margin or nonlinear decision boundaries, but they are primarily predictive and do not provide a transparent criterion-wise explanation of why one alternative is preferred over another (Herbrich et al., 2000; Han et al., 2012; Claesen et al., 2014; Fürnkranz et al., 2014). By contrast, linear utility, TOPSIS, ideal-solution distance, and Tchebycheff-style formulations preserve interpretability but remain limited in their ability to represent non-monotonic desirability and learned compensation behavior at the same time (Hwang & Yoon, 1981; Stanimirovic et al., n.d.; Chakraborty et al., 2023; Lin et al., 2024). Attitudinal Choquet models move closer to this goal by incorporating interaction and attitude, yet they still do not directly model the criterion-level sweet-spot structure that is central to the present framework (Aggarwal, 2018; Aggarwal & Fallah Tehrani, 2019). The proposed model differs from these baselines by combining interpretable local desirability shaping, learned criterion importance, and a learned global aggregation attitude within one preference-learning pipeline.

Table 3: Research Gap Map: Prior Limitations and How the Proposed Method Addresses Them

Targeting gap	Where prior work falls short	Representative prior lines	What this proposal adds	Why it matters
“Monotonic utility” is unrealistic for many criteria	“Bigger is better / smaller is better” cannot express <i>sweet-spot</i> preferences (too low and too high both undesirable)	Many MAUT / linear utility baselines; standard additive/weighted averaging	Parametric bell-shaped desirability per criterion peaking at an <i>optimal</i> value	Prevents ranking an extreme option (e.g., cheapest) above a balanced option when DM wants moderation
Optimal value should be choice-set sensitive	Fixed “target” or fixed utility curves ignore that what looks “optimal” depends on the available alternatives	Point-elicited interpolation; fixed utility templates	Optimal point derived from the criterion’s distribution over the choice-set	Mimics realistic shopping/selection: “best” depends on what’s on the shelf

Targeting gap	Where prior work falls short	Representative prior lines	What this proposal adds	Why it matters
DM compensation behavior must be modeled explicitly	Additive / weighted sum assumes a single compensation style (high score can always offset a weak criterion)	Additive, weighted averaging, many learning-to-rank models	Aggregation designed to reflect DM's compensation attitude (less compensatory → penalize imbalance)	Captures “deal-breaker” behavior: one weak criterion can't be fully rescued by another
Balanced alternatives should be rewarded over extreme ones	Distance/linear models can still over-reward extremes depending on weights/scaling	Linear utility; ideal-solution distance; norms	Bell-shaped desirability + compensation-aware aggregation jointly prefer “well-rounded” profiles	Aligns with common human preference: avoid “cheap but unusable” or “premium but wasteful” extremes
Preference learning should be interpretable, not just predictive	RankSVM / kernels predict well but don't expose <i>why</i> in MCDA terms (criterion desirabilities + tradeoff attitude)	RankSVM; RBF kernel	Explicit criterion-level desirability curves and interpretable parameters	Lets you explain decisions to stakeholders (and audit DM preference assumptions)
Validation should go beyond MCDA narrative	MCDA papers often stop at ranking; predictive generalization isn't checked	Pure MCDA utility/OWA stories	Validation inside an ML framework (treat DM choice prediction as a learning problem)	Tests whether the preference model generalizes rather than overfitting the observed set
Criteria can be conflicting and closely inter-related	Simple additive frameworks assume separability; interaction models can be heavy or not tied	Additive/weighted; Choquet/ACI (interaction)	A compact desirability + aggregation design intended to remain practical while handling conflict	Targets the middle ground: more realistic than monotonic MAUT, lighter than full interaction learning

Targeting gap	Where prior work falls short	Representative prior lines	What this proposal adds	Why it matters
	to “perceived feature value” framing		via “balancedness”	
Baselines may not capture “perceived feature value” shaping	Even sophisticated baselines (e.g., ACI) may emphasize interaction but not the “sweet spot” desirability per criterion	Attitudinal Choquet; distance to ideal	Desirability shaping is central (criterion-wise) rather than an afterthought	If the DM’s primary behavior is “prefer the middle,” interaction modeling alone won’t fix it

Table 3 consolidates the specific gaps that motivate the proposed framework. The key distinction of the present model is not merely that it aggregates multiple criteria, but that it does so after transforming each criterion through a generalized-Gaussian desirability function centered on a standardized reference region derived from the current choice set. This design allows the model to represent moderation-seeking behavior that monotonic MAUT, fixed-target ideal-solution methods, and purely predictive rankers do not explicitly encode. In addition, the use of compensative weighted averaging with a learned attitude parameter separates criterion importance from compensation style, while Bradley–Terry likelihood estimation places the MCDA formulation inside a predictive ranking setting rather than leaving it as a descriptive scoring rule. Taken together, these features position the proposed approach between classical monotonic utility models and more complex interaction-heavy formulations: it is more behaviorally expressive than weighted-sum or distance-based baselines, yet more compact and interpretable than richer interaction models.

3. Methodology

3.1 Project objective and study design

This project implements an end-to-end preference-learning multi-criteria ranking pipeline that ranks a set of alternatives described by multiple conflicting criteria (Figure 2). The core idea is to replace standard monotonic utilities (“more is always better”) with a non-monotonic

desirability model that peaks around a standardized reference region and penalizes deviations on either side, then learn how criteria should be combined using a compensatory aggregation that can behave in a more AND-like or OR-like manner depending on a learned attitude parameter. Figure 1 presents a high-level conceptual overview of the proposed preference-learning MCDA framework, showing the transformation from raw criterion values to criterion-wise desirability and finally to the aggregated utility used for ranking.

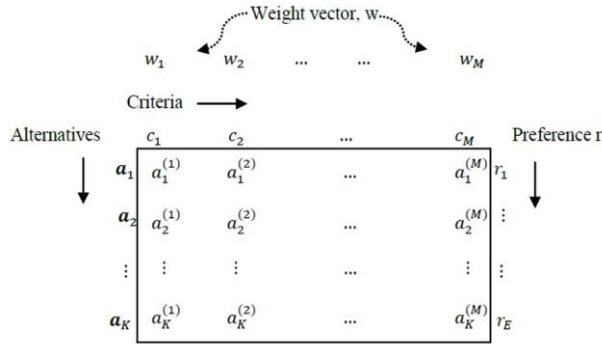


Figure 1: System overview of the preference-learning desirability model.

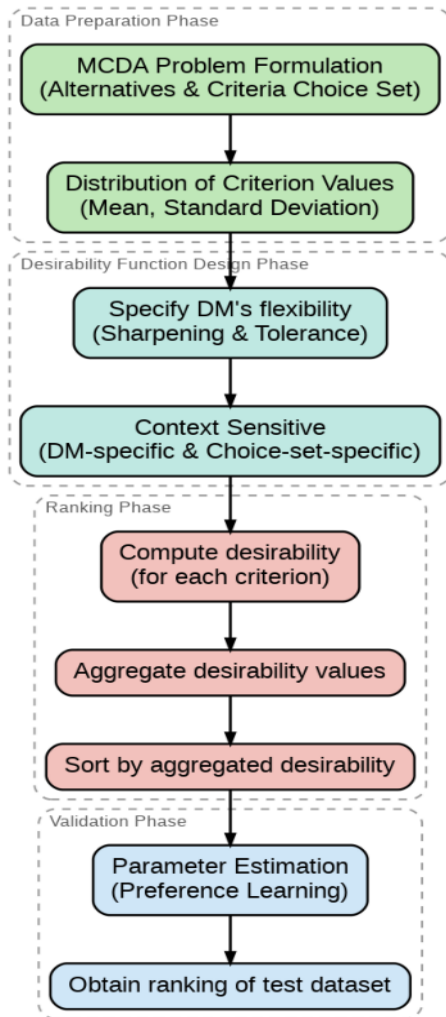


Figure 2: Flow diagram of the proposed methodology

For clarity, the complete experimental and learning procedure adopted in this study is summarized in Algorithm 1. This algorithm consolidates the main steps of the proposed framework before the individual components are described in detail in the following subsections.

Algorithm 1: End-to-end preference-learning MCDA ranking procedure with generalized-Gaussian desirability, CWA aggregation, and Bradley–Terry estimation.

Input: $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N, M, T, \rho, K$

Output: $\{\alpha^{(t)}, w^{(t)}, \lambda^{(t)}, \text{Acc}^{(t)}\}_{t=1}^T$, mean $\pm$ std accuracy

Steps:

1. Preprocess $\mathcal{D}$: remove non-informative fields, encode categorical criteria, handle missing values.
2. Construct ordinal labels r_i :
 - continuous $y_i \rightarrow$ quantile-based ranks;
 - ordinal $y_i \rightarrow$ ordered class levels.
3. For $t = 1, \dots, T$:

- Split $\mathcal{D} \rightarrow \mathcal{D}_{\text{tr}}^{(t)}, \mathcal{D}_{\text{te}}^{(t)}$ with ratio ρ .
- Standardize x using $\mathcal{D}_{\text{tr}}^{(t)}$ statistics.
- Sample K pairs (i, j) from $\mathcal{D}_{\text{tr}}^{(t)}$ such that $r_i > r_j$.
- Initialize unconstrained variables for α, w, λ .
- Parameterize:

$$\alpha = \text{softplus}(\tilde{\alpha}), w = \text{softmax}(\tilde{w}), \lambda = \phi(\tilde{\lambda})$$

- Compute desirability:

$$g_{im} = \exp(-|a_i^{(m)}|^{\alpha^{(m)}})$$

- Compute information values I_{im} .
- Aggregate by CWA:

$$U_i = \text{CWA}(I_i; w, \lambda)$$

- Bradley–Terry probability:

$$P(i > j) = \sigma(U_i - U_j)$$

- Minimize

$$\mathcal{L}(\alpha, w, \lambda) = - \sum_{(i,j)} \log \sigma(U_i - U_j)$$

using L-BFGS-B.

- Score test alternatives: $U_i, i \in \mathcal{D}_{\text{te}}^{(t)}$.
- Compute pairwise ranking accuracy:

$$\text{Acc}^{(t)} = \frac{\#\{(i,j): r_i \neq r_j, (U_i - U_j)(r_i - r_j) > 0\}}{\#\{(i,j): r_i \neq r_j\}}$$

- Store $\alpha^{(t)}, w^{(t)}, \lambda^{(t)}, \text{Acc}^{(t)}$.

4. Compute

$$\bar{\text{Acc}}, \text{Std}(\text{Acc})$$

over $t = 1, \dots, T$.

5. Repeat Steps 3–4 under:

- MAUT: $\lambda = 1, g = 1$
- No-desirability: $g = 1$
- Fixed-uniform weights: $w_m = 1/M$

6. Report full-model and ablation/baseline results.

3.2 Dataset, cleaning, and target ranking labels

Three UCI benchmark datasets (FRANK, 2010) were used to cover both regression-derived rankings and naturally ordinal labels: Auto MPG (398 instances, 7 features), Car Evaluation (1728 instances, 6 features), and Computer Hardware (209 instances, 9 features). In Auto MPG, a mixture of numeric criteria and non-numeric identifiers was provided; non-informative fields such as the car name were removed, only modeling criteria were retained, and records with missing values were discarded. In Car Evaluation, all predictors were categorical, so categorical encoding (e.g., one-hot or ordinal encoding per attribute) was applied and category levels were verified for consistency. In Computer Hardware, integer-valued system descriptors were given; numeric criteria were retained, inputs were standardized for scale comparability,

and any missing entries were handled under the same row-discard protocol. For a consistent supervised ordinal ranking target, continuous outcomes in Auto MPG and Computer Hardware were converted into ordinal preference labels by discretizing the target values into six quantile-based rank levels, with higher MPG or higher measured performance mapped to higher ranks. In contrast, the Car Evaluation acceptability label was already ordinal in nature, so class categories were mapped directly to an ordered rank scale from least to most desirable without discretization.

3.3 Feature preprocessing and scaling

Let $X \in \mathbb{R}^{K \times M}$ denote the raw criterion matrix for K alternatives and M criteria. **Z-score** standardization is implemented.

The standardized matrix A is computed per criterion m as:

$$a_i^{(m)} = \frac{x_i^{(m)} - \mu_x^{(m)}}{\sigma_x^{(m)}}$$

where $\mu_x^{(m)}$ and $\sigma_x^{(m)}$ are the empirical mean and standard deviation of criterion m in the current split.

3.4 Criterion-level desirability (non-monotonic local utility)

For each criterion m , desirability is modeled by a generalized Gaussian bell-shaped function defined on the standardized criterion $a_i^{(m)}$. Because $a_i^{(m)}$ is z-scored within each split, the desirability function is centered at the standardized reference point 0 with unit scale. For alternative i ,

$$g_i^{(m)} = \exp\left(-|a_i^{(m)}|^{\alpha^{(m)}}\right)$$

where $\alpha^{(m)} > 0$ controls how sharply deviations from the ideal region are penalized; larger $\alpha^{(m)}$ yields narrower acceptance around the ideal.

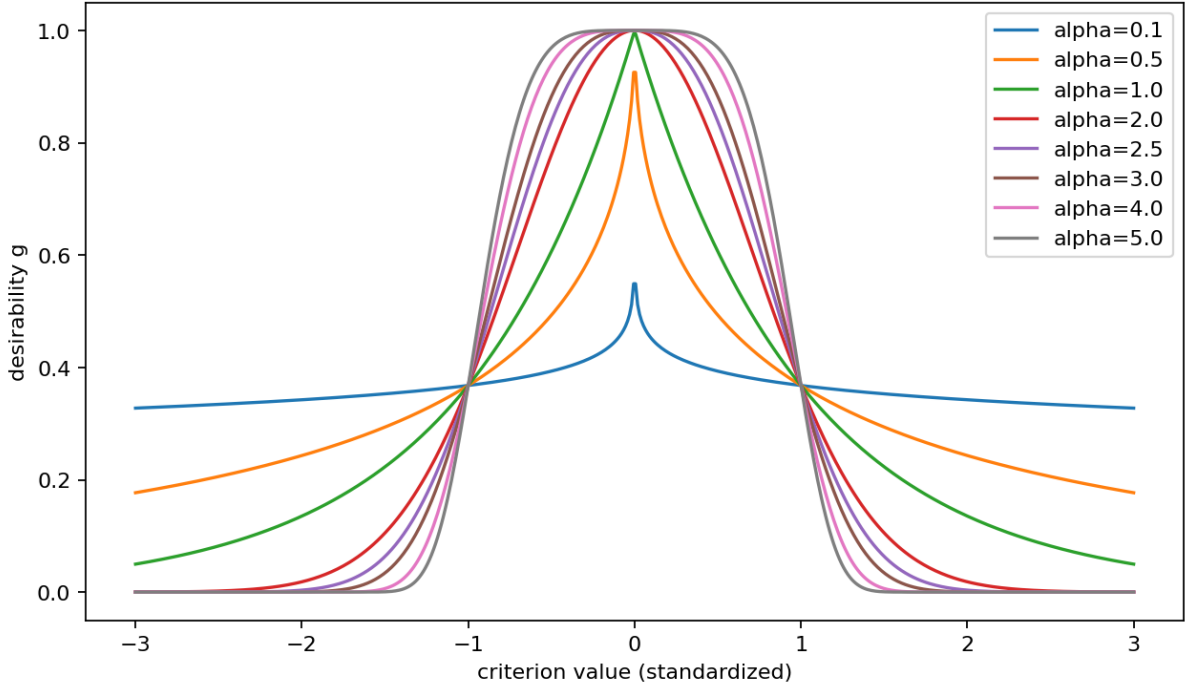


Figure 3: Generalized Gaussian desirability curves (standardized scale).

3.5 Information value per criterion

To ensure that a criterion contributes strongly only when it is both influential in magnitude and close to its standardized reference region, an information value is computed:

$$S_i^{(m)} = a_i^{(m)} \cdot g_i^{(m)}$$

This gating mechanism suppresses extreme criterion values that are far from the standardized reference region, thereby capturing situations in which extreme trade-offs are less desirable.

3.6 Global aggregation with compensative weighted averaging (CWA)

Criterion-wise information values are aggregated into a single score G_i for each alternative using Compensative Weighted Averaging (CWA):

$$G_i = \frac{\log \left(\sum_{m=1}^M w_m \exp \left(\log(\lambda) S_i^{(m)} \right) \right)}{\log(\lambda)}$$

Here, $w_m \geq 0$ are criterion weights with $\sum_m w_m = 1$, and $\lambda > 0$ is an attitude/compensation parameter:

- $\lambda \approx 1$: reduces to a weighted arithmetic mean (neutral compensation).
- $\lambda > 1$: more OR-like (emphasizes standout criteria).
- $\lambda < 1$: more AND-like (penalizes weak criteria more strongly).

A numerically stable limit is used when λ approaches 1.

3.7 Preference generation for learning-to-rank

From the ordinal labels in the training split, we generate a set of pairwise preferences. A preference pair (i, j) is valid when $y_i > y_j$, meaning alternative i should be ranked above j . For each randomized trial, we sample $N = 600$ such pairs by selecting i uniformly and then selecting j uniformly from instances with strictly lower rank than i . These pairwise constraints form the supervision signal for maximum likelihood training.

3.8 Parameter learning via Bradley–Terry likelihood (maximum likelihood estimation)

Given utilities G_i , the probability that i is preferred over j follows a Bradley–Terry model:

$$P(i \succ j) = \sigma(G_i - G_j),$$

where $\sigma(\cdot)$ is the logistic sigmoid. The parameters

$$P = \alpha^{(1 \dots M)}, w_{1 \dots M}, \lambda$$

are learned by minimizing the negative log-likelihood over sampled preference pairs:

$$\mathcal{L} = \sum_{(i,j)} \log \left(1 + \exp \left(-(G_i - G_j) \right) \right)$$

Constraint handling. To satisfy constraints without manual projection:

- $\alpha^{(m)}$ is parameterized using softplus to ensure $\alpha^{(m)} > 0$,
- w is parameterized using softmax to ensure $w_m \geq 0$ and $\sum_m w_m = 1$,
- λ is parameterized as $\lambda = \exp(\eta)$ to enforce $\lambda > 0$, with η bounded to prevent overflow.

Optimization uses L-BFGS-B for stability and efficiency.

3.9 Evaluation metric: pairwise ranking accuracy

Performance is evaluated on the held-out test split using pairwise ranking accuracy. For all test pairs (p, q) with different ground-truth ranks, a pair is counted correct if:

$$(y_p > y_q) \Leftrightarrow (G_p > G_q)$$

Accuracy is computed as concordant pairs divided by all comparable pairs. To avoid $O(n^2)$ cost, discordant pairs are counted efficiently via an inversion-count procedure (Fenwick tree / BIT) after mapping predicted scores to ranks.

3.10 Experimental protocol (100 randomized trials)

To ensure robustness and reduce split bias, the full procedure is repeated for 100 random seeds:

1. Randomly split data into 50% train / 50% test.
2. Sample 600 preference pairs from training labels.
3. Fit the proposed model by maximum likelihood.
4. Score test alternatives and compute pairwise accuracy.
5. Report mean and standard deviation across trials.

4. Results

The proposed preference-learning desirability model was evaluated on three UCI choice sets to cover both regression-derived ordinal ranks and naturally ordinal labels: Auto MPG, Car Evaluation, and Computer Hardware. In Auto MPG and Computer Hardware, continuous targets were discretized into six quantile-based ordinal rank levels, with higher target values mapped to higher ranks; in Car Evaluation, the original acceptability categories were mapped directly to an ordered rank scale. In each of 100 randomized trials, a 50/50 train–test split was formed, 600 preference pairs were sampled from the training labels, parameters $\{\alpha^{(m)}\}_{m=1}^M$, w , and λ were fit by maximum-likelihood optimization, and pairwise ranking accuracy was computed on the test set.

Across 100 randomized trials, dataset-wise mean pairwise ranking accuracies were obtained under the FULL configuration (learn α, w, λ). In Auto MPG, an accuracy of 0.8190 ± 0.0141

was obtained; in Car Evaluation, an accuracy of 0.8958 ± 0.0097 was obtained; and in Computer Hardware, an accuracy of 0.8508 ± 0.0292 was obtained. A unimodal accuracy distribution was observed for each dataset across the 100 trials (Figure 4), and trial-to-trial fluctuations remained modest.

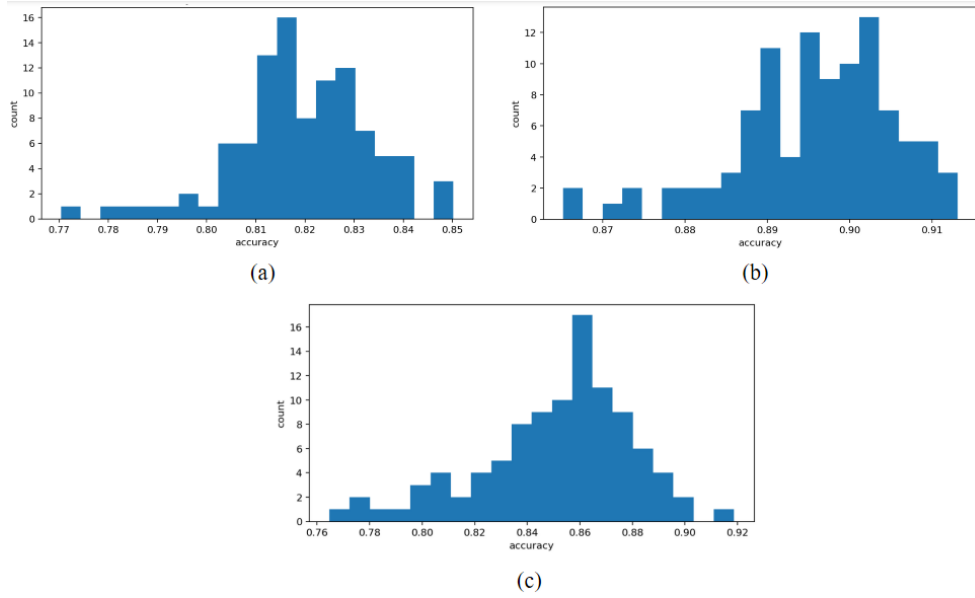


Figure 4: Accuracy distribution over 100 iterations of all the datasets.

Figure 5 shows that accuracy variability across trials was concentrated around the corresponding means, indicating consistent performance under split variability and preference-pair sampling.

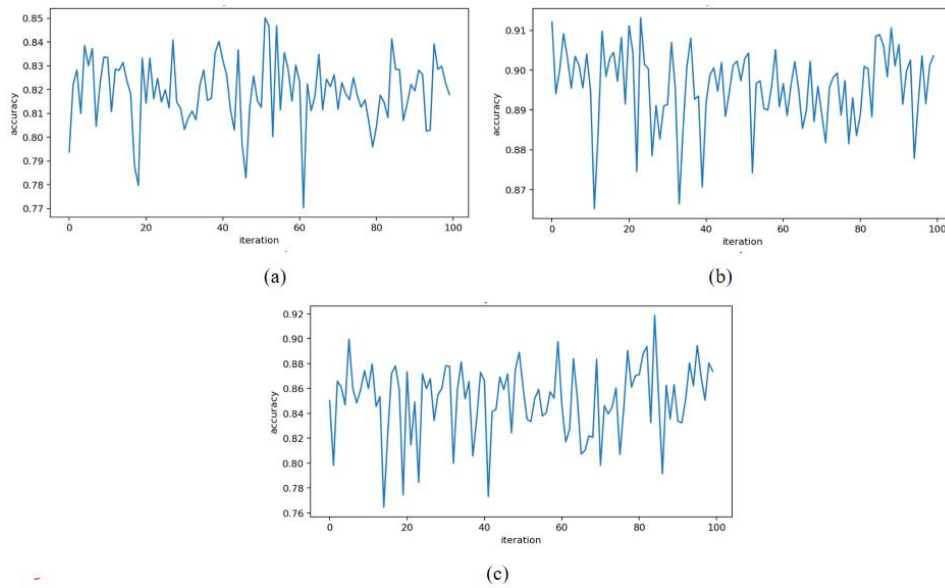


Figure 5: Accuracy over 100 iterations of all the datasets.

Representative training traces were characterized by a monotonic decrease in negative log-likelihood (Figure 6), suggesting stable convergence under the L-BFGS-B optimizer.

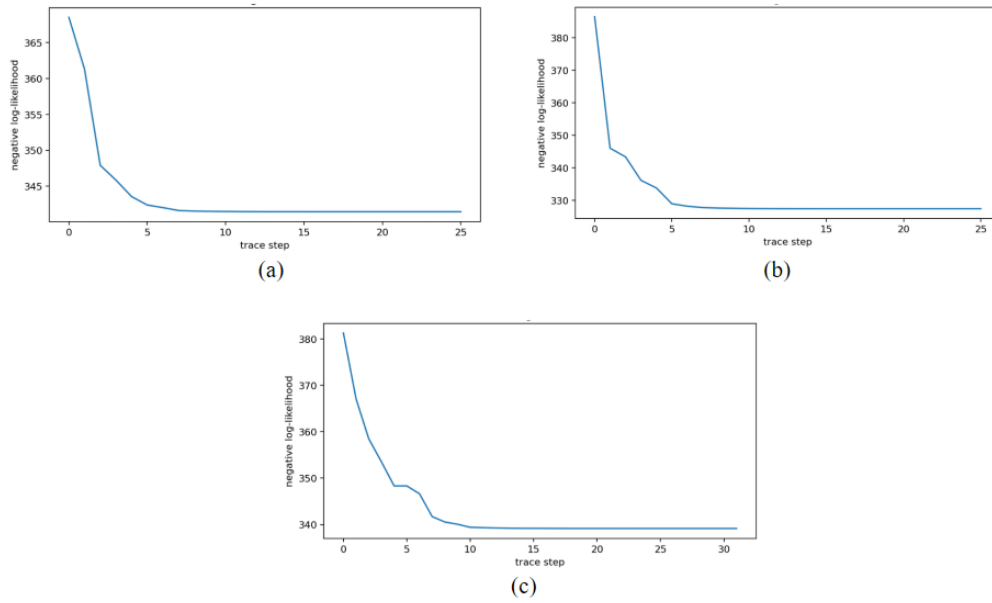


Figure 6: Training loss trace of all the datasets.

The learned λ values were concentrated in the $\lambda \gg 1$ regime across trials indicating an OR-like aggregation tendency under the sampled supervision signal.

The mean learned weights (Figure 7) are non-uniform, indicating that the ranking is driven by a subset of criteria rather than equal contribution across dimensions.

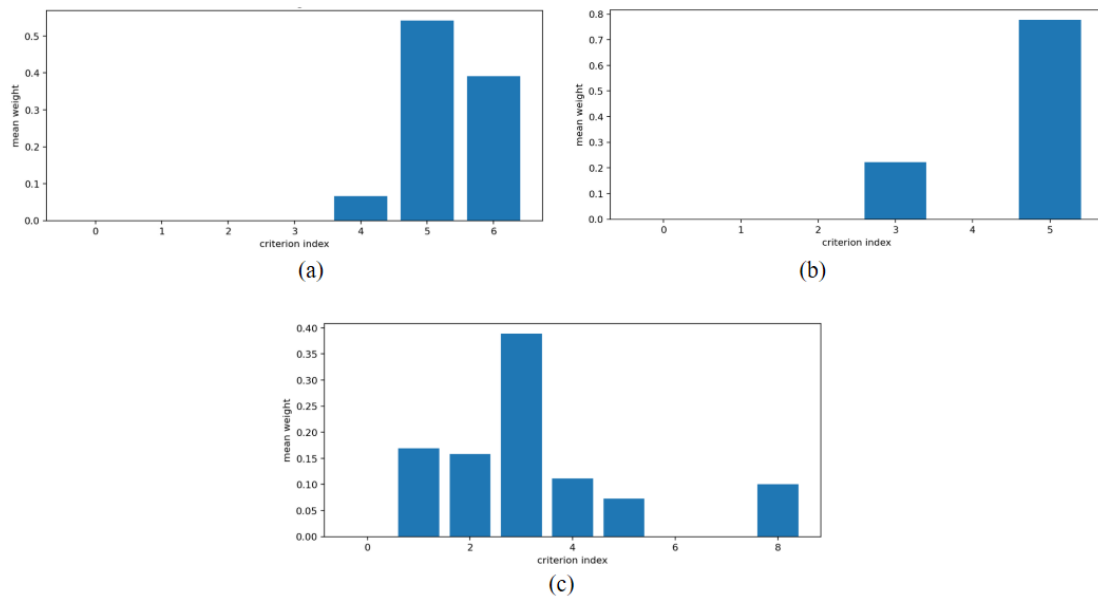


Figure 7: Mean learned weights across 100 iterations of all the datasets.

The mean α values (Figure 8) vary across criteria, reflecting different tolerance levels for deviations from the choice-set-defined “ideal” region.

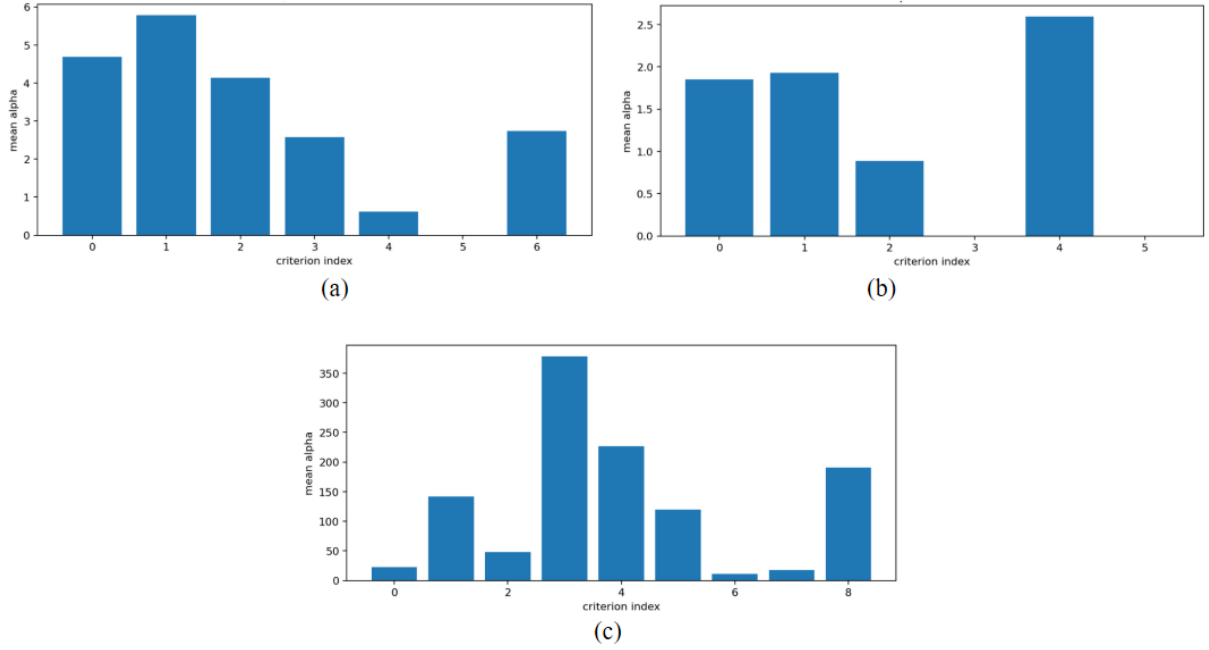


Figure 8: Mean learned alphas across 100 iterations of all the datasets.

Finally, the learned values were frequently concentrated near the upper bound of the imposed range, which is consistent with an OR-like aggregation tendency in which strong performance on key criteria can dominate the global utility.

5. Discussion

Classical weighted-sum scoring implicitly assumes monotonic utility: improving any criterion always increases preference. In real multi-criteria decisions, this is often unrealistic because extreme values can represent imbalance (e.g., overly optimizing one attribute while neglecting others). The implemented model addresses this by coupling three components: (i) bell-shaped desirability per criterion, which rewards values near a choice-set-defined “ideal range,” (ii) information value $S_i^{(m)} = a_i^{(m)} g_i^{(m)}$, which suppresses extreme-but-undesirable criterion values, and (iii) compensation-aware aggregation via the CWA operator controlled by λ . Together, these mechanisms let the model prefer alternatives that are strong on important criteria without blindly rewarding extremes.

A key interpretability output is the learned λ , which determines whether aggregation behaves more AND-like ($\lambda < 1$, weak criteria dominate), mean-like ($\lambda \approx 1$), or OR-like ($\lambda > 1$,

standout criteria can dominate). In the experiments, λ was parameterized as $\lambda = \exp(\eta)$ and η was bounded within $[\eta_{\min}, \eta_{\max}]$ to prevent numerical overflow. The bound $\eta_{\max} = 10$ (equivalently $\lambda_{\max} = e^{\eta_{\max}}$) was imposed and was occasionally attained, indicating a strongly OR-like aggregation tendency under the given supervision signal. Practically, this means the learned ranking model can rank an alternative highly if it performs exceptionally well on a subset of criteria, even if other criteria are only moderate. This behavior is plausible given the proxy-label construction (MPG-driven ranking): if one or two attributes strongly correlate with MPG in the dataset, the likelihood objective will push λ toward an OR-like regime that amplifies standout contributions.

Important implication: λ hitting the upper bound is not “just a parameter value”—it signals that under the current supervision signal, the model prefers high compensation. This is a meaningful learned behavior, but it also suggests the need for baseline/ablation checks to confirm it is not an artifact of (i) discretized labels or (ii) limited preference-pair sampling.

The learned weights w represent criterion importance, while α represents tolerance for deviating from the ideal range. This separation is valuable: a criterion can be heavily weighted yet tolerant (high w , low α), or less central but strict about staying near its ideal (moderate w , high α). This is a key advantage over monotonic models where “importance” must be expressed only through weights and cannot encode “avoid extremes” behavior.

A limitation is that the “decision maker” is defined implicitly by quantile discretization of MPG, i.e., preferences reflect “higher MPG is better” rather than a human multi-criteria trade-off. This introduces two effects:

- Boundary noise: near quantile cutoffs, very similar MPG values can map to different ranks, creating ambiguous preference pairs and lowering achievable accuracy.
- Proxy preference: the learned desirability/aggregation explains MPG-driven ordering, not necessarily a real-world multi-criteria decision process.

Therefore, the results should be interpreted as showing that the method can learn an interpretable ranking model consistent with an ordinal target derived from a real dataset, rather than as definitive evidence of human decision behavior. To further assess where the observed ranking performance was coming from, ablation and baseline comparisons were conducted by selectively removing desirability shaping, compensation control, or learned criterion weighting.

Table 4: Mean Pairwise Ranking Accuracy (mean $\pm$ std) for the Full Model and Ablation/Baseline Variants

Operation	mean $\pm$ std		
	Auto MPG	Car Evaluation	Computer Hardware
FULL (learn α , w , λ)	0.8190 $\pm$ 0.0141	0.8958 $\pm$ 0.0097	0.8508 $\pm$ 0.0292
MAUT baseline ($\lambda=1$, $g=1$, learn w)	0.8307 $\pm$ 0.0132	0.8647 $\pm$ 0.0157	0.8481 $\pm$ 0.0167
No-desirability ($g=1$, learn w , λ)	0.8206 $\pm$ 0.0139	0.8450 $\pm$ 0.0072	0.8259 $\pm$ 0.0185
Fixed-uniform w ($w=1/M$, learn α , λ)	0.6524 $\pm$ 0.0383	0.7393 $\pm$ 0.0134	0.7546 $\pm$ 0.0516

A slight reversal was observed on Auto MPG, where the MAUT baseline marginally exceeded the FULL configuration. This outcome was expected because the supervision signal in Auto MPG was induced directly from a single monotonic target (MPG), for which monotonic aggregation is naturally aligned with the proxy ranking labels. In contrast, stronger benefits from desirability shaping and compensation control were expected to emerge when ordinal labels reflect broader multi-attribute acceptability patterns, as suggested by the higher FULL performance on Car Evaluation.

6. Conclusion

In this project, an end-to-end MCDA-inspired preference-learning ranking pipeline was implemented to examine a core limitation of monotonic utility models in multi-criteria ranking. Standard “more is better” assumptions were replaced by a non-monotonic, bell-shaped desirability formulation that rewards criterion values near an ideal region and penalizes deviations on either side. By defining desirability on standardized criteria and introducing a sharpenable parameter α , a context-sensitive and interpretable representation of criterion-level preference was obtained. In addition, compensative weighted averaging was used to aggregate criterion-wise information values into global utilities, and the attitude parameter λ was learned to control whether aggregation behaved in a more AND-like, mean-like, or OR-like manner. A Bradley–Terry likelihood was then used to estimate (α, w, λ) from sampled pairwise

preferences, enabling the MCDA model to be validated under a predictive learning-to-rank framework rather than being presented only as a descriptive ranking heuristic.

Empirical validation was conducted on three UCI choice sets spanning both regression-derived ordinal ranks and naturally ordinal labels: Auto MPG, Car Evaluation, and Computer Hardware. Across 100 randomized trials, mean pairwise ranking accuracies of 0.8190 ± 0.0141 , 0.8958 ± 0.0097 , and 0.8508 ± 0.0292 were obtained under the FULL configuration, respectively, indicating that stable generalization was achieved under repeated resampling. Convergence behavior was characterized by monotonic decreases in negative log-likelihood, and parameter distributions were found to be interpretable: λ was concentrated in the $\lambda \gg 1$ regime and occasionally reached the imposed bound $\eta_{\max} = 10$, suggesting that an OR-like compensation tendency was learned under the observed supervision signal.

Ablation results further clarified where performance was being generated. It was shown that removing desirability shaping ($g \equiv 1$) reduced accuracy across datasets, and fixing weights to uniform values produced a substantial degradation, indicating that both criterion-level shaping and learned importance weights were contributing materially to ranking quality. A slight reversal was observed on Auto MPG where a monotonic MAUT baseline marginally exceeded the FULL configuration, and this outcome was consistent with the fact that Auto MPG supervision was induced from a single monotonic target (MPG), for which monotonic aggregation can be naturally aligned with the proxy ranking labels. In contrast, stronger benefits from desirability shaping and compensation control were reflected on Car Evaluation, where ordinal acceptability labels more directly represent multi-attribute preference structure.

Overall, an interpretable preference-learning MCDA model was demonstrated in which (i) criterion-level desirability could be shaped to avoid extreme, unbalanced alternatives and (ii) aggregation attitude could be learned from data rather than being fixed a priori. At the same time, limitations were introduced by proxy-label construction through quantile discretization, which can create boundary noise and restrict conclusions to consistency with an induced ordinal target rather than definitive human decision behavior. As a result, the primary conclusion supported by the experiments was that an explainable desirability-and-compensation model can be fit reliably to ordinal preference signals and can generalize across held-out comparisons, while exposing meaningful parameters (α, w, λ) that describe tolerance and compensation behavior in a decision-analytic form.

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